
EE/CprE/SE 492 WEEKLY REPORT 2

September 15, 2025 - September 18, 2025

Group number: 16

Project title: Outdoor Roomba

Client &/Advisor: Prof. Diane Rover

Team Members/Role:

Brodie M Bates - Embedded system

Daniel Vergara Pinilla - Machine Learning

Mitchell D Owen - Project Tester

Om Shukla - Frontend Developer

Sudipta Halder - Simulation developer

○ Weekly Summary

This week, we will test our motors and batteries, where we found both were faulty and had developed damage over the summer break. Therefore are planning to use the motors and batteries from a project called SCUTTLE Robot, an open-source robotics platform originating in a basement lab at Texas A&M. The SCUTTLE Robot claims to be able to have a 40kg payload. Because of the claims and prices, we felt it would be best for our project, so we started to buy the parts and have most of them arrive.

We have started writing drivers for the Lidar sensor, magnetic encoders, and motor drivers. We are using EfficientDet-Lite0 as our Machine Learning model, and it can successfully detect objects like bottles, cans, and humans(person), but still needs more training data since it is still unreliable. Our website has been updated and is ready to connect to the servers at ETG.

○ Past week accomplishments

- Sudipta Halder: Looked into Scuttle Robot as a backup for our motors and battery. Bought the motors, batteries, and other components (timing belts, screws etc). Continued to 3D print the mounts needed.
- Brodie Bates: Tested the sensor to make sure it works with drivers made by LDROBOT. Then, started programming an ESP32 to read the LiDAR sensor data and report it through its serial connection. Programmed arduino uno to run the motor controllers to test.
- Daniel Pinilla: debugged segmentation error on the implementation of the new camera module in the ML code, which was preventing the camera to work as expected
- Om Shukla: Finished converting the HTML code to this REACT app with a mock server implementation. I have also started working on sending requests to the backend server for information about users and their login information, so I can use actual accounts to log in to the dashboard.
- Mitchell Owen: This week went over the calculations for the motor gear and to the motor controller to get an estimate of specs to make sure it will be a valuable component to our robot. A photo of the specs spreadsheet is attached above.

This calculation spreadsheet is important to get valuable points for our robot. The spreadsheet shows the type speed we can have with our current motor gear. If we need a new one we can implement the new parameters in this spreadsheet instead of guessing which one will be good for what we need. This spreadsheet showed that we might need 4 motor gears instead of 2 due to the expected speed and climbable slope we will need to accomplish.

Parameter	Value (enter inputs here)	Units / Notes
Robot Mass	7	kg (total mass incl. battery) We need max weight with cans.
Wheel Radius	0.05	m (half diameter)
Motor Rated Torque (per motor)	0.2157463	N-m (2.2 * 0.0980665) 2.2 is the rated torque in kg.cm
Motor Speed (output shaft)	200	RPM
Number of Motors (driving wheels)	2	
Gravity (g)	9.81	m/s^2
Calculated Result		
Torque per motor (N-m)	0.2157463	
Force per motor at wheel rim (N)	4.314926	Torque / Wheel radius
Total traction force (N) (all motors)	8.629852	(Motor Rated Torque / (Wheel Radius * Number of motors)
Wheel linear speed (m/s)	1.047197551	$v = (RPM/60) * 2\pi * \text{wheel radius}$
Mechanical power per motor (W)	4.518579941	torque * angular speed (rad/s)
Total mechanical power (W) (all motors)	9.037159882	
Theoretical acceleration (m/s^2)	1.232836	(Motor Rated Torque / Wheel Radius * Number of motors) / Mass
Maximum climbable slope (deg) (ideal)	7.219527371	$\text{deg} = \text{asin}(F_total / (m * g))$

- **Pending issues**

- Brodie Bates: I'm currently having trouble with the LiDAR sensor looking into scuttlebots documentation to find a solution.
- Sudipta Halder: all of the parts for Scuttle bot needed for our project have arrived, but the shipping time for the timing belts is being delayed.

○ **Individual contributions**

<u>NAME</u>	<u>Individual Contributions</u>	<u>Hours this week</u>	<u>HOURS cumulative</u>
Brodie Bates	Test LiDAR sensor. Programming ESP32. Testing motor controllers.	7	9
Om Shukla	Finished REACT app, started backend connection implementation.	10	12
Sudipta Halder	Buying parts for scuttlerobot robot BoM to replace our motors, batteries, wheels and other components. Cad and 3D printed models needed for	20	25
Daniel Mauricio Vergara	worked in debugging segmentation error in the Raspberry Pi due to the error while implementing the new camera in the environment of the ML code	10	12
Mitchell Owen	Calculation for the power of the motor gear and controller	7	9

○ **Plans for the upcoming week**

- Sudipta Halder: Assemble the motors and base with the parts from Scuttle Robot. Test our current Microcontroller system. Start implementing our design and move on to testing.
- Brodie Bates: Get the ESP32 or other microcontroller to read the LiDAR sensor data and output it. Get motor controllers working and functions to move forward and back a specific distance.
- Om Shukla: Work more on the backend connection implementation. I also need to modify the backend code to add certain endpoints that I will be using for user login.
- Mitchell Owen: Test the motor gear with our finished base model to confirm the specs that were acquired through my spreadsheet.
- Daniel Pinilla: Improve dataset for ML since the YOLO became unreliable for the project. More testing and pictures required to successfully train the model to understand correctly the objects and fulfill the requirements

- **Summary of weekly advisor meeting**

We have had two advisor meetings since the last report. In these meetings, we discussed demoing every 3 weeks to help push us to progress. Also, we talked about creating an interface between parts, even if it is just a mock-up, and using simulation tools to help test out the robots and our AI code.

In the second meeting, we discussed having a regular meeting with all team members as well as going over our Gantt chart for the first semester and making changes as needed. It was also mentioned that we should be prioritizing tasks and choosing week by week what tasks need to be done. In the last part of the meeting, we talked about having multiple users in our app, monitoring the performance of our main CPU Raspberry Pi, and testing our multiprocessor architecture to ensure it is optimal for our design.